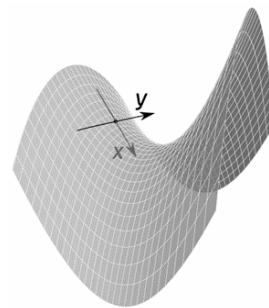


MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.2 Vectors

Multivariable Calculus requires us to not just know where we are, but also where we are going. Therefore, we introduce a new tool, a *vector*!

Drawn as an arrow, this mathematical object conveys both direction and magnitude, allowing us to explore familiar Calculus 1 concepts such as *motion* and *slope*.



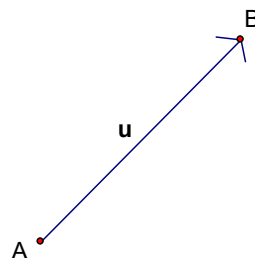
Definitions:

- A scalar is a number x . It tells you *how much* of something you have. Speed, distance, mass, and temperature are all scalar quantities. They don't care where you're going, just how much you've got.
- A vector is an object $\vec{u}$ that has both magnitude and direction. A vector is often represented by an directed line segment (an arrow): its length tells you its magnitude and the way it points tells you its direction.

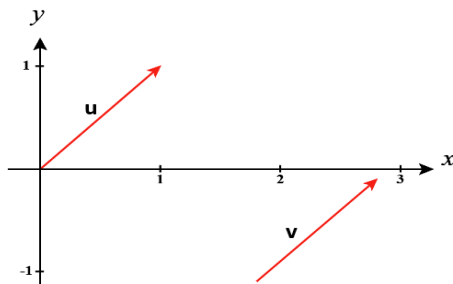
Note: The definition of a vector above works for how we will think about vectors in MTH 202. However, the general definition of vector is quite different from the definition above.

Terminology and Notation:

- Vectors are labeled by either using vector notation ($\vec{u}$), boldface ($\mathbf{u}$), or endpoint notation ($\overrightarrow{AB}$).
- The point A is called the initial point (or tail) of the vector $\mathbf{u}$.
- The point B is called the terminal point (or tip) of the vector $\mathbf{u}$.
- One can write $\mathbf{u} = \overrightarrow{AB}$



Note: One of the biggest subtleties involving vectors is that vectors are equivalent (or equal) as long as they have the same length and direction. For example, in the picture below $\mathbf{u} = \mathbf{v}$. This means that vectors do not have inherent initial or terminal points.



Definition: The zero vector denoted by $\mathbf{0}$ is the vector with length zero, and it is the only vector without a specified direction.

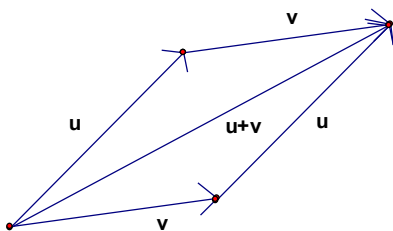
Definition: Suppose that to get from the initial point of vector $\mathbf{a}$ to the terminal point of vector $\mathbf{a}$ we must move a_1 units in the x -direction, a_2 units in the y -direction, and a_3 units in the z -direction. We write $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, and a_1 , a_2 , and a_3 are called the components of vector $\mathbf{a}$ (we use this same idea in two dimensions as well).

Note: The same definition can be applied to vectors that exists in 2-space, 3-space, and beyond. We can have vectors of that can exist in n -dimensional space.

Thinking About Vectors Operations Geometrically:

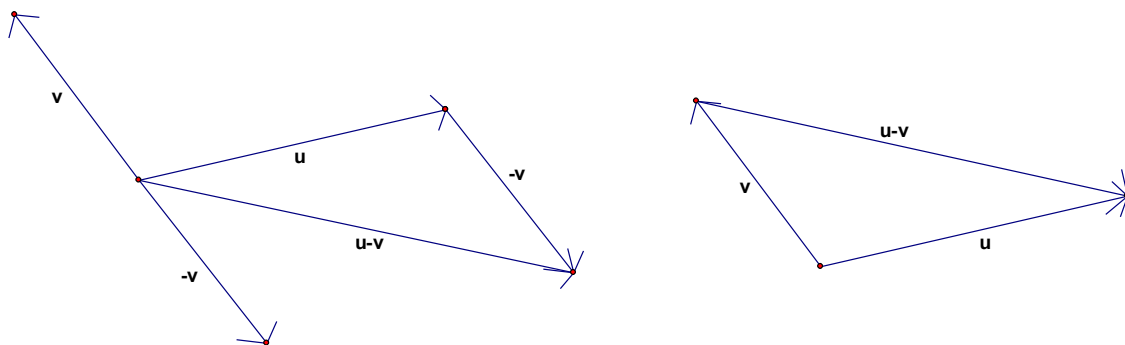
Definition: If $\mathbf{u}$ and $\mathbf{v}$ are vectors positioned so the initial point of $\mathbf{v}$ is at the terminal point of $\mathbf{u}$, then the sum $\mathbf{u} + \mathbf{v}$ is the vector from the initial point of $\mathbf{u}$ to the terminal point of $\mathbf{v}$.

Note: An informal way to say this definition is that we add vectors tip to tail



Definition: If c is a scalar and $\mathbf{v}$ is a vector then the scalar multiple $c\mathbf{v}$ is the vector whose length is $|c|$ times the length of $\mathbf{v}$ and whose direction is the same as $\mathbf{v}$ if $c > 0$ and is opposite to $\mathbf{v}$ if $c < 0$. If $c = 0$ or $\mathbf{v} = \mathbf{0}$, then $c\mathbf{v} = \mathbf{0}$.

Remark: Now that we have defined both vector addition and scalar multiplication there are two different ways we can think about drawing $\mathbf{u} - \mathbf{v}$:



Note: The left picture uses the idea of scalar multiplication, and the right picture uses the idea that $\mathbf{v} + (\mathbf{u} - \mathbf{v}) = \mathbf{u}$.

Thinking About Vectors Operations Algebraically:

When working in 2D:

- Vector Addition: $\langle a_1, a_2 \rangle + \langle b_1, b_2 \rangle = \langle a_1 + b_1, a_2 + b_2 \rangle$
- Vector Subtraction: $\langle a_1, a_2 \rangle - \langle b_1, b_2 \rangle = \langle a_1 - b_1, a_2 - b_2 \rangle$
- Scalar Multiplication: $c \langle a_1, a_2 \rangle = \langle ca_1, ca_2 \rangle$

When working in 3D:

- Vector Addition: $\langle a_1, a_2, a_3 \rangle + \langle b_1, b_2, b_3 \rangle = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$
- Vector Subtraction: $\langle a_1, a_2, a_3 \rangle - \langle b_1, b_2, b_3 \rangle = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$
- Scalar Multiplication: $c \langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$

Definition: Suppose $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2 \rangle$. The length (or sometimes called norm) of $\mathbf{a}$ is written $|\mathbf{a}|$ or $\|\mathbf{a}\|$, and it is given by $\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$. Similarly, $\|\mathbf{b}\| = \sqrt{b_1^2 + b_2^2}$.

Note: Length is a scalar!!!!!!

Example 1: Suppose that $\mathbf{a} = \langle 3, 4 \rangle$ and $\mathbf{b} = \langle 5, 12 \rangle$. Find $\|\mathbf{a}\|\mathbf{a} + \|\mathbf{b}\|\mathbf{b}$.

We note that $\|\mathbf{a}\|\mathbf{a} + \|\mathbf{b}\|\mathbf{b} = \sqrt{3^2 + 4^2} \langle 3, 4 \rangle + \sqrt{5^2 + 12^2} \langle 5, 12 \rangle = 5 \langle 3, 4 \rangle + 13 \langle 5, 12 \rangle = \langle 15, 20 \rangle + \langle 65, 156 \rangle = \langle 80, 176 \rangle$.

Definition: The vector with its initial point as the origin and its terminal point as some point P is the position vector of the point P . If the point P has coordinates (a_1, a_2, a_3) we represent the position vector of the point P as: $\mathbf{a} = \overrightarrow{OP} = \langle a_1, a_2, a_3 \rangle$

Note: This definition applies to vectors in any dimension.

Example 2: Suppose the vector $\mathbf{u}$ in 2-space has initial point $(4, 4)$ and terminal point $(8, 5)$.

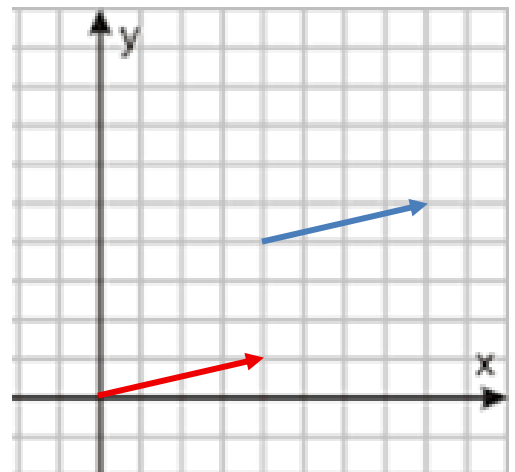
a) Find the algebraic form of the vector $\mathbf{u}$. Sketch this vector.

In order to get from $(4, 4)$ to $(8, 5)$ we must move 4 units in the x direction and 1 unit in the y direction. Therefore,
 $\mathbf{u} = \langle 4, 1 \rangle$

We could also just express this algebraically as
 $\mathbf{u} = \langle 8 - 4, 5 - 4 \rangle = \langle 4, 1 \rangle$
(Shown in blue)

b) Sketch the position vector for the point $(4, 1)$.

(Shown in red) Note this vector is the same as $\mathbf{u}$.



Example 3: Suppose the vector $\mathbf{v}$ in 3-space has initial point $(2, -3, 4)$ and terminal point $(-2, 1, 1)$.

a) Find an algebraic form of the vector $\mathbf{v}$.

In order to get from $(2, -3, 4)$ to $(-2, 1, 1)$ we must move -4 units in the x direction, 4 units in the y direction, and -3 units in the z direction. So, the answer is $\mathbf{a} = \langle -4, 4, -3 \rangle$.

We could also just express this algebraically as

$$\mathbf{v} = \langle -2 - 2, 1 - (-3), 1 - 4 \rangle = \langle -4, 4, -3 \rangle$$

b) State the algebraic form of the position vector for the point $(-4, 4, -3)$.

This is still $\mathbf{v} = \langle -4, 4, -3 \rangle$. Remember that vectors are *location independent*!

Definition: We let V_2 denote the set of all two-dimensional vectors and V_3 denote the set of all three-dimensional vectors. More generally, we let V_n denote the set of all n -dimensional vectors.

Properties of Vectors: Suppose that $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors in V_n , and suppose c and d are scalars. Then, we have:

- | | |
|--|---|
| 1. $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ | 5. $c(\mathbf{a} + \mathbf{b}) = c\mathbf{a} + c\mathbf{b}$ |
| 2. $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$ | 6. $(c + d)\mathbf{a} = c\mathbf{a} + d\mathbf{a}$ |
| 3. $\mathbf{a} + \mathbf{0} = \mathbf{a}$ | 7. $(cd)\mathbf{a} = c(d\mathbf{a})$ |
| 4. $\mathbf{a} + (-\mathbf{a}) = \mathbf{0}$ | 8. $1\mathbf{a} = \mathbf{a}$ |

Definition: The standard basis vectors are the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$. These vectors are given by the formulas: $\mathbf{i} = \langle 1, 0, 0 \rangle$, $\mathbf{j} = \langle 0, 1, 0 \rangle$, and $\mathbf{k} = \langle 0, 0, 1 \rangle$.

We now have two different ways of representing vectors in three space. For example, $\langle 1, -2, 6 \rangle = \mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$. Similarly in two dimensions we can write $\langle 1, -2 \rangle = \mathbf{i} - 2\mathbf{j}$ (in this context we are taking $\mathbf{i} = \langle 1, 0 \rangle$ and $\mathbf{j} = \langle 0, 1 \rangle$).

Example 4: Suppose that $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} - 2\mathbf{k}$

a) Show that vector $\mathbf{a}$ is equivalent to the vector $\langle 1, -2, 6 \rangle$.

$$\begin{aligned}\mathbf{a} = \mathbf{i} - 2\mathbf{j} + 6\mathbf{k} &= \langle 1, 0, 0 \rangle - 2\langle 0, 1, 0 \rangle + 6\langle 0, 0, 1 \rangle = \langle 1, 0, 0 \rangle - \langle 0, 2, 0 \rangle + \langle 0, 0, 6 \rangle = \langle 1 - 0 + 0, 0 - 2 + 0, 0 - 0 + 6 \rangle \\ &= \langle 1, -2, 6 \rangle\end{aligned}$$

b) Find $2\mathbf{a} - \mathbf{b}$ in terms of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.

We calculate that $2\mathbf{a} - \mathbf{b} = 2(\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}) - (3\mathbf{i} - 2\mathbf{k}) = 2\mathbf{i} - 4\mathbf{j} + 12\mathbf{k} - 3\mathbf{i} + 2\mathbf{k} = -\mathbf{i} - 4\mathbf{j} + 14\mathbf{k}$.

Definitions:

- Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel if they are scalar multiples of each other. Mathematically, $\mathbf{u}$ and $\mathbf{v}$ are parallel if there exists a scalar c such that $\mathbf{u} = c\mathbf{v}$.
- A unit vector is a vector whose length is 1.

Note: The standard basis vectors are all unit vectors.

Example 5: Suppose that $\mathbf{a} = \langle 2, 6, -1 \rangle$ and $\mathbf{b} = \langle -6, -18, 3 \rangle$.

a) Are vectors $\mathbf{a}$ and $\mathbf{b}$ parallel? Justify your answer.

These vectors are parallel by definition since $\mathbf{b} = -3\mathbf{a}$.

b) Find a unit vector which is parallel to $\mathbf{a}$.

One way to do this is to note that $\|\mathbf{a}\| = \sqrt{2^2 + 6^2 + (-1)^2} = \sqrt{41}$. Then the vector

$\frac{1}{\|\mathbf{a}\|}\mathbf{a} = \left\langle \frac{2}{\sqrt{41}}, \frac{6}{\sqrt{41}}, \frac{-1}{\sqrt{41}} \right\rangle$ has length 1 and is parallel to $\mathbf{a}$ by definition.

Remark: If $\mathbf{a}$ is a nonzero vector, then the unit vector that has the same direction as $\mathbf{a}$ is $\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|}$